朗城科技
LONGGUARD TECHNOLOGY

广工大数控装备协同创新研究院
智能防护技术研发中心

Passive Upper Limb Exoskeleton

LC Power— U/A

Product Overview

Pure mechanical bionic structure for precise directional assistance. Lightweight, reliable, adaptable to complex environments. Multi-level adjustable to fit different body types. Ideal for automotive assembly, power maintenance, construction, aircraft repair, agricultural picking. Reduces muscle load, improves efficiency and comfort.

Key Features

Lightweight

Carbon fiber + Oxford cloth, lightweight, ergonomic, no movement restriction.

Passive Operation

No power supply, zero energy consumption, no outage risk, safe & stable.

High Efficiency

Elastic energy storage, supports up to 8kg load, reduces strain, boosts endurance.

Durable

Minimalist structure, impact-resistant, wear-resistant, low maintenance.

4-gear adjustment, assistance activates instantly when the arm is raised.

Quick-release structure for easy donning/doffing.

Carbon fiber back bar fits various heights.

Quick-release armrest for instant escape.

Adjustable waist for different body shapes.

Specification

Passive Upper Limb Exoskeleton

Basic Information			
Model	LC Power-U/P	**Color**	Black
Users	Workers with prolonged arm-raising tasks		
Scenarios	Automotive assembly, power maintenance, construction, etc.		

Hardware Parameters			
Materials	Carbon fiber + aluminum alloy	**Capacity**	Max. 8kg
Locking Mode	Semi-automatic	**Assistance Gears**	3 gears
Assistance/Unlocking Angles	50°, 70°, 90°, 110°/ 115°		

Fabric Details	
Outer Fabric	Wear-resistant Oxford fabric
Lining Fabric	Breathable 3D mesh fabric

Other Information			
Size	Standard Size / Plus Size	**Net Weight**	<2kg
Package	50*40*12CM	**Optional Accessories**	Neck support
Warranty	1 Year	**Service Number**	15989175854

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Flexible Waist Exoskeleton

LC Flex — W

Product Overview

This lightweight, unpowered lumbar exoskeleton is designed for frequent bending tasks. It stores and releases energy via elastic material deformation, acting as a "smart belt" to deliver upward assistive thrust during bending. Significantly reduces lumbar strain and injury risk.

Key Features

Lightweight

Light materials plus minimalist design for comfortable, sweat-free all day.

Body-Fitted

Conforms to body contours, non-restrictive for full freedom of movement.

Strain Reduction

Precisely aids bending and standing motions, effectively reducing lumbar muscle load.

Lumbar Support

Distributes lumbar pressure to prevent strain, offering protection for standing and heavy labor.

- Adjustable back fits multiple body types
- No rigid compression for free movement
- Thickened lumbar padding for pressure distribution
- Passive all-mechanical structure for safe use
- Cross-traction assistance structure for stable load reduction

Specification

Flexible Waist Exoskeleton

Basic Information					
Model	LC Flex-W	**Color**	Black	**Users**	Manual laborers
Scenarios	Industrial logistics, medical elderly care and other frequent bending scenarios				

Structural Parameters			
Assistance Type	Non-powered	**Applicable Height**	160-190cm
Performance Index	Assistance Range: 0-30kg (dynamically adjustable)		
Energy Storage	Elastic polyester latex fiber		
Core Structure	Integrated back-lumbar support bracket + bilateral thigh linkage strap system		

Fabric Details	
Outer Fabric	Wear-resistant Oxford fabric
Lining Fabric	Breathable 3D mesh fabric

Other Information			
Size	Standard Size / Plus Size	**Net Weight**	≤1kg
Package	40*30*10CM	**Service Number**	15989175854
Warranty	1 Year		

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Wrist Repetitive Work Assessment Glove

LC Gloves-01

Product Overview

Designed for manufacturing jobs with repetitive forceful wrist movements, this device complies with GBZ 336-2025 *Diagnostic Criteria for Occupational Carpal Tunnel Syndrome*.With dual-posture sensors and edge-AI computing, it tracks wrist angles, valid work motions, operation frequency, duration and load intensity in real time. It supports wireless data upload and tablet-based analysis, delivering accurate data for occupational health monitoring, ergonomic assessment and workstation optimization to evaluate wrist musculoskeletal exposure risks.

Key Features

Precision Sensor

Dual-point attitude sensing eliminates whole-arm-movement interference and accurately captures wrist angles.

Real-Time Processing

Onboard glove computing for angle calculation, motion counting and data statistics. No external device needed.

Wireless Data Sync

Wireless sync to PC for real-time angle curves and work reports. Data storable, exportable and traceable.

Stable & Reliable

Vibration and interference resistant for high-intensity environments in electronics, automotive, assembly and furniture manufacturing.

- Half-finger design fits multiple hand sizes
- Gloves fit snugly without slack
- Lightweight and responsive
- Adjustable wristband for various sizes

Specification

Wrist Repetitive Work Assessment Glove

Basic Information			
Model	LC Gloves-01	**Color**	Grey
Scenarios	Occupational-disease prevention hospitals (on-site occupational health inspection services)/ Third-party occupational health testing & evaluation institutions/ Manufacturers for ergonomic workstation assessment, etc.		

Measurement Performance			
Angle Range	±180°	**Recognition Threshold**	15° (customizable)
Static Accuracy		≤±0.1°	
Frequency Counting	Automatically counts ≥4 motions per minute		
Work Duration	Daily working hour accumulation		
Measurement Dimensions	Wrist flexion/extension, radial/ulnar deviation angles		

Other Information			
Size	M, L, XL etc.	**Net Weight**	200g
Package	23*17*8CM	**Service Number**	15989175854
Warranty	1 Year	**Optional Accessories**	Tablet PC / Action Camera

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PCM Cooling Vest

LC Frost — PCM/A

Product Overview

Built with high-energy-density Phase Change Material (PCM) that absorbs heat at a constant temperature (~23°C). Delivers consistent comfort cooling. One cold storage cycle provides 1-2 hours of cooling at 35°C. Reusable thousands of times – a wearable "mini air conditioner" solution.

Key Features

Mild Cooling

Phase change temperature (22–28°C) aligns with human comfort zone. Gentle, non-irritating cooling avoids discomfort or cold risk from sudden temperature drops.

Passive Cooling

All-physical, power-free cooling with no electricity or water needed. Portable and wearable. No wires, noise or energy consumption, unrestricted movement.

Stable Performance

C17 n-alkane PCM delivers stable, long-lasting cooling with no significant degradation. Consistent performance for reliable cooling.

Eco-Friendly

Plant/biomass-based C17 n-alkane PCM (non-petroleum derived). Biodegradable and environmentally friendly.

Eco-friendly cooling technology for fast cooling

Adjustable waist for all body types

TPU outer layer for easy cleaning

Simple to wear, highly versatile for matching

Specification

PCM Cooling Vest

Basic Information			
Model	LC Frost - PCM/A	**Color**	Transparent (customizable)
Users	Workers in high-temperature environments		
Scenarios	High-temperature environments		

PCM Parameters			
Heat of Fusion/(J/g)	225	**Phase Change Temperature**	22.08℃
Cooling Duration	1~2 hours (reference based on labor intensity and ambient temperature)		
Cold Storage Duration	Frozen at -18°C for 1 hour, refrigerated at 0-4°C for 1-2 hours		

Fabric Details	
Material	TPU

Other Information			
Size	One size	**Net Weight**	0.55kg
Package	40*30*10CM	**Service Number**	15989175854
Warranty	1 Year	**Optional Accessories**	PCM Color

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Flame Retardant PCM Cooling Vest

LC Frost — PCM/G

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Product Overview

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Passive Cooling

All-physical, power-free cooling with no electricity or water needed. Portable and wearable. No wires, noise or energy consumption, unrestricted movement.

Stable Performance

C17 n-alkane PCM delivers stable, long-lasting cooling with no significant degradation. Consistent performance for reliable cooling.

Eco-Friendly

Plant/biomass-based C17 n-alkane PCM (non-petroleum derived). Biodegradable and environmentally friendly.

Wear-resistant outer fabric

Elastic magic tape waistband fits all body shapes

Eco-friendly, fast cooling

Safety warning reflective strips

Specification

PCM Cooling Vest

Basic Information			
Model	LC Frost - PCM/G	**Color**	Grey (customizable)
Users	Workers in high-temperature environments		
Scenarios	High-temperature environments		

PCM Parameters			
Heat of Fusion/(J/g)	225	**Phase Change Temperature**	22.08℃
Cooling Duration	1~2 hours (reference based on labor intensity and ambient temperature)		
Cold Storage Duration	Frozen at -18°C for 1 hour, refrigerated at 0-4°C for 1-2 hours		

Fabric Details	
Material	Antistatic and flame retardant fabric

Other Information			
Size	Standard Size / Plus Size	**Net Weight**	0.96kg
Package	40*30*10CM	**Service Number**	15989175854
Warranty	1 Year		

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Semiconductor Water Cooling Vest

LC Frost — TEC/W

Product Overview

This semiconductor water-cooled vest is an active wearable cooling device. Powered by thermoelectric cooling chips and water circulation system, it delivers steady, adjustable torso cooling for staff working in high-temperature conditions. It applies to firefighting, public security, power, construction, sanitation and other industries with extreme heat exposure.

Key Features

Accurate Temp Control

Instant cooling after startup. Intelligent host supports multi-level temperature adjustment to keep body comfortably cool, no passive cooling limits. Optional phase change strips boost cooling efficiency.

Breathable

Built-in quiet rear fan dissipates back heat fast; chest fan brings facial cooling, optimizing air circulation.

Durable Battery

High-capacity battery sustains 2-4 hours continuous use, fits long-time intense work. Light design ensures free movement and comfort.

Safe Performance

Fully enclosed water circuit adapts to tough working scenarios with stable and secure operation.

Anti-static material, soft and comfortable

Adjustable waistband fits all body shapes

Adjustable temperature

Circulating air supply to keep dry

Specification

Semiconductor Water Cooling Vest

Basic Information			
Model	LC Frost - TEC/W	**Color**	Grey (customizable)
Users	Workers in high-temperature environments		
Scenarios	High-temperature environments		

Semiconductor Parameters			
Cooling Capacity	80W	**Rated Voltage**	24V
Rated Power	100W	**Cooling Response Time**	Stable in 2min
Single Battery Life	1~2 hours (reference based on labor intensity and ambient temperature)		

Fabric Details	
Outer Fabric	Antistatic and flame retardant fabric
Lining Fabric	TPU

Other Information			
Size	One size	**Net Weight**	<3kg
Package	40*30*10CM	**Service Number**	15989175854
Warranty	1 Year	**Optional Accessories**	PCM packs

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Direct-Cooling Semiconductor Vest

LC Frost— TEC/D

Product Overview

This direct-cooling semiconductor cooling vest is active personal cooling gear leveraging the Peltier effect. Equipped with high-performance TEC modules, it requires no refrigerant or compressor. Direct-contact heat absorption and rapid heat dissipation deliver instant cooling and intelligent temperature regulation.
Ideal for high-temperature work, medical protection and outdoor use.

Key Features

Direct Cooling

Requires no ice or liquid coolant. Semiconductor cooling chips deliver direct cooling immediately upon power-on, offering fast response and precise temperature control.

Low Noise

No compressor operation noise, only low-noise cooling fans with almost zero vibration, suitable for quiet operation scenarios.

Long Battery Life

Matched with high-capacity special batteries, it can work continuously for 2-8 hours, meeting the single operation duration of most high-intensity services.

Eco-friendly

No refrigerant required, simple and convenient, no pipeline leakage risk; cooling chip life exceeds 20,000 hours, almost no daily maintenance required.

- Instant Cooling without Pre-cooling
- Tailored to fit the body shape, unrestricted operation activities
- Low-noise cooling fans, no noise or stuffiness after long wearing
- Modular detachable and washable, breathable fabric

Specification

Direct-Cooling Semiconductor Vest

Basic Information			
Model	LC Frost - TEC/D	**Color**	Black
Users	Workers in high-temperature environments		
Scenarios	High-temperature environments		

Semiconductor Parameters			
Cooling Capacity	45W	**Rated Voltage**	12V
Rated Power	60W	**Cooling Response Time**	Stable in 2s
Single Battery Life	2-8 hours (single battery lasts 2 hours)		

Fabric Details	
Outer Fabric	Antistatic and flame retardant fabric

Other Information			
Size	Standard Size / Plus Size	**Net Weight**	~1.7kg
Package	40*30*10CM	**Service Number**	15989175854
Warranty	1 Year		

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Compressor Water Cooling Vest

LC Frost — ACC/W

Product Overview

Langcheng compressor water-cooled vest applies micro inverter compressor and liquid circulation refrigeration. Its closed-loop system circulates low-temperature water to evenly dissipate body heat. Featuring high cooling power, wide temperature range and stable constant temperature, it serves as professional high-temperature protective cooling gear.

Key Features

Active Cooling

Instant cooling on startup. Multi-level temperature adjustment maintains comfortable body temperature. Optional ice crystals further boost cooling performance

Lightweight

Built-in slim cooling structure weighs under 4kg, ensuring unrestricted motion and comfy wear.

Long Endurance

High-capacity battery supports 2-4 hours continuous operation, suited for long intensive tasks.

Stable Performance

Fully sealed water circuit prevents leakage. Waterproof and shockproof for harsh working conditions.

Anti-static material, soft and comfortable

Adjustable waistband fits all body shapes

Adjustable temperature

Circulating air supply to keep dry

Specification

Compressor Water Cooling Vest

Basic Information			
Model	LC Frost - ACC/W	**Color**	Grey
Users	Workers in high-temperature environments		
Scenarios	High-temperature environments		

Compressor Parameters			
Cooling Capacity	60-300W	**Rated Voltage**	24V
Rated Power	110W	**Cooling Response Time**	Stable in 3m
Single Battery Life	1~2 hours (reference based on labor intensity and ambient temperature)		

Fabric Details	
Outer Fabric	Antistatic and flame retardant fabric
Lining Fabric	TPU

Other Information			
Size	One size	**Net Weight**	<4kg
Package	40*30*10CM	**Service Number**	15989175854
Warranty	1 Year		

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